

TRC-8 Tracker Controller

Operations, Position Feedback & Fault Response Manual

DOCUMENT	TRC-OM-008
VERSION	2.7
EFFECTIVE	May 14, 2026
STATUS	Approved — Synthetic
MANUFACTURE R	TrackRight Controls (Synthetic)

6. Position Monitoring

The TRC-8 compares commanded row angle with measured position feedback and records absolute deviation during automatic tracking.

6.1 Signal interpretation

COMMANDED ANGLE	Controller-requested row angle; it does not prove physical row position.
MEASURED ANGLE	Reported position; it may be affected by sensing, linkage, or drive conditions.
POSITION DEVIATION	Absolute difference; it identifies disagreement, not the failed component.

6.2 Motion safety boundary

- Keep the affected row in Safe Stow Hold while position disagreement remains unresolved.
- Remain outside the tracker movement envelope and use the site-approved motion-control process.
- Contact with drive, torque-tube, linkage, or sensor components requires qualified maintenance and the applicable isolation and lockout/tagout procedure.

Safety note: A stationary row may move unexpectedly if controls, stored energy, or an obstruction change. This synthetic manual does not replace site procedures.

6.3 Fault P09 — Tracker Position Deviation

Fault P09 is asserted when absolute commanded-versus-measured row-position deviation exceeds 5.0 degrees during the controller diagnostic window. The affected row enters Safe Stow Hold and automatic motion is inhibited pending inspection.

Expected behavior: P09 establishes that commanded and measured position disagreed beyond the configured limit. It does not, by itself, prove a failed actuator, position sensor, drive, or linkage. Obstruction, row interference, mechanical drag, feedback sensing, and control-path conditions remain possible until inspected.

Required response

- 1 Confirm the asset, P09 event window, and Safe Stow Hold with the control room.
- 2 Review commanded angle, measured angle, position deviation, and drive-motor current for the same window.
- 3 From outside the movement envelope, inspect the visible row path and structure for vegetation, debris, row-to-row interference, displaced modules, or damaged members.
- 4 Do not command motion, enter the movement envelope, or touch drive components under this diagnostic step.
- 5 Use the approved site inspection procedure and return-to-automatic checklist before release.

Evidence interpretation: Elevated drive current is consistent with increased loading, but neither current nor position deviation alone establishes the underlying cause.